

PREDICT, DETECT & OPTIMIZE

RecallNext

**Find the missing fact that
changes the recall
decision**

Evidence-optimized food recall investigation with Exasol

INVESTIGATION BRIEF

**From incomplete
movement records to a
controlled next action.**

TEAM- 4DUMBS

- Bhavishya
- Adya Mohanka
- D Harini
- Sakthivel A

A recall notice does not identify every downstream shipment

The notice starts the investigation. Private movement records determine downstream scope.

OPENFDA CONTEXT

H-1259-2026

Class I

Undeclared peanuts

Product · hazard · affected codes · recalling firm ·
classification · distribution summary

MISSING
PRIVATE
MAPPING

DISTRIBUTOR QUESTION

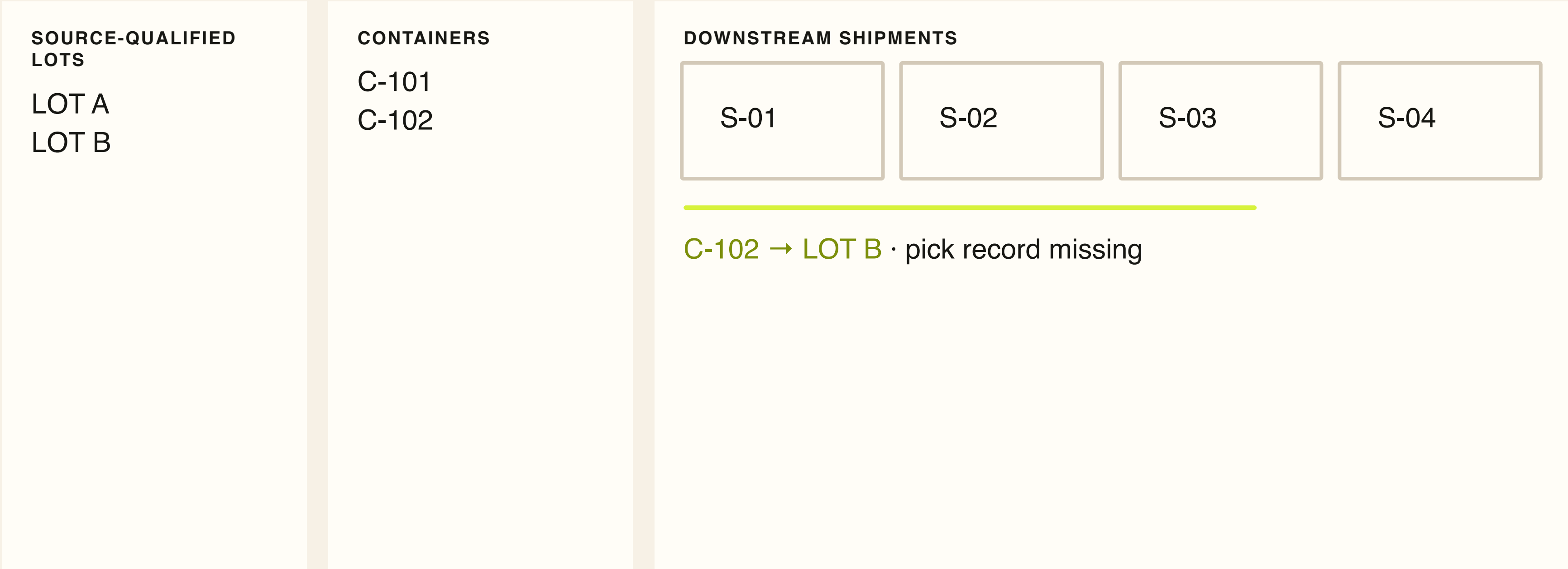
Where did the lot go?

Which containers received the lot?
Which cases entered each shipment?
Which customers received them?

Public recall context is real. Downstream movement records remain private to the distributor.

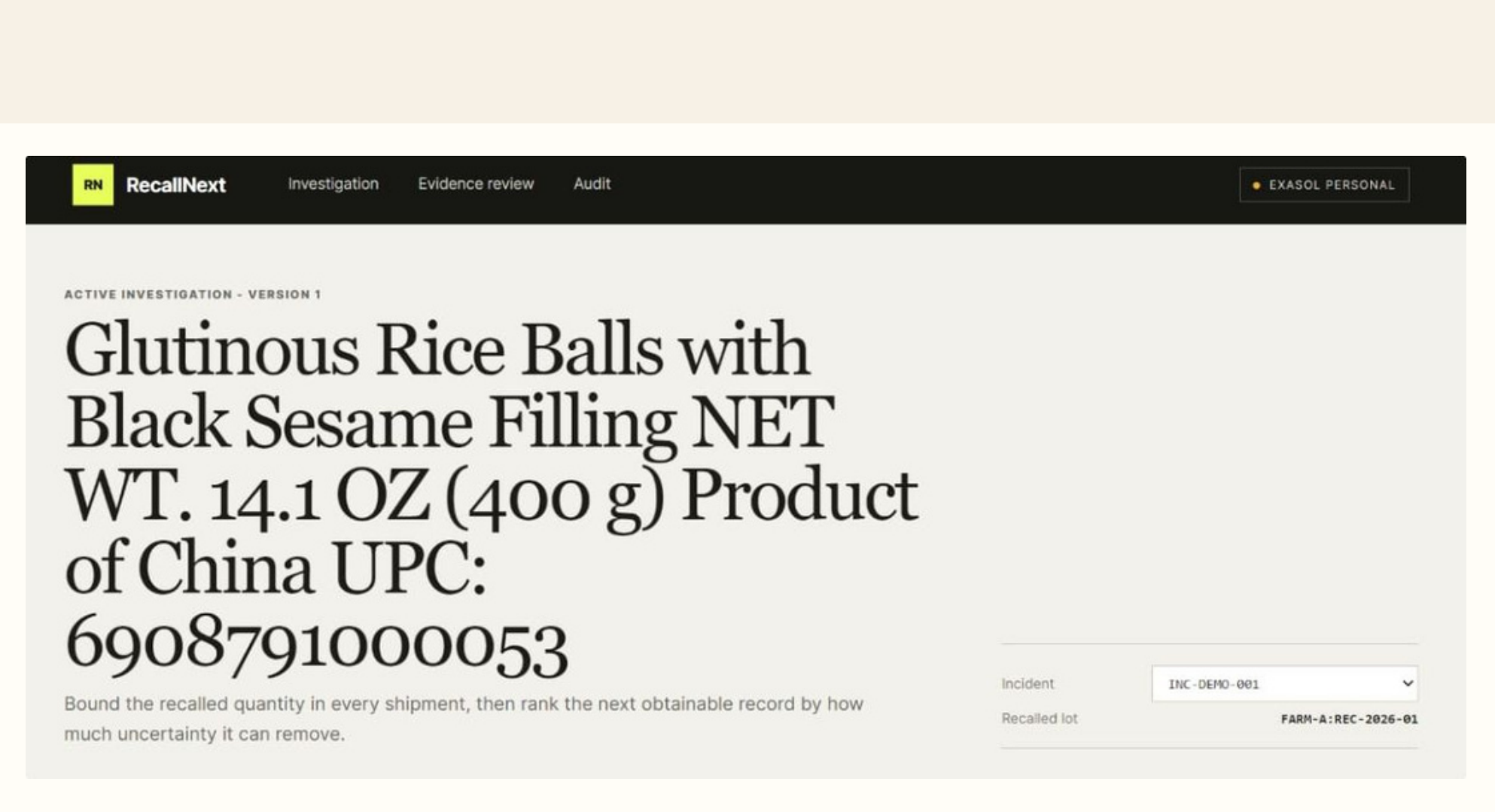
Incomplete lot tracking creates a decision problem

The missing fact is operational: hold everything, or exclude stock without enough evidence?



Several allocations remain consistent with the records. The right response is neither an indefinite hold nor a guessed history.

RecallNext converts uncertainty into an investigation plan



CURRENT UNCERTAINTY
What remains possible across the records.

NEXT RECORD TO CHECK
Which obtainable evidence could narrow the decision.

Preserve every complete lot-to-shipment history supported by current records; calculate minimum and maximum recalled cases possible in each shipment. Simulate obtainable labels, scans, pick logs and manifests, then rank the next evidence action by decision value and estimated effort. Require human acceptance before any decision changes; record every version, conflict and retraction in Exasol.

Six operational problems have explicit controls

Operational problem	RecallNext control
Incomplete lot-to-shipment mapping	Preserve all feasible histories
Exposure differs across histories	Calculate exposure bounds
Arbitrary record chasing	Rank obtainable actions
Missing coverage or conflicts	Return UNRESOLVED
Unsupported AI or user fact	Pending until human verifies
Withdrawn evidence	Rebuild and create a new version

Controls keep uncertainty explicit instead of silently converting assumptions into decisions.

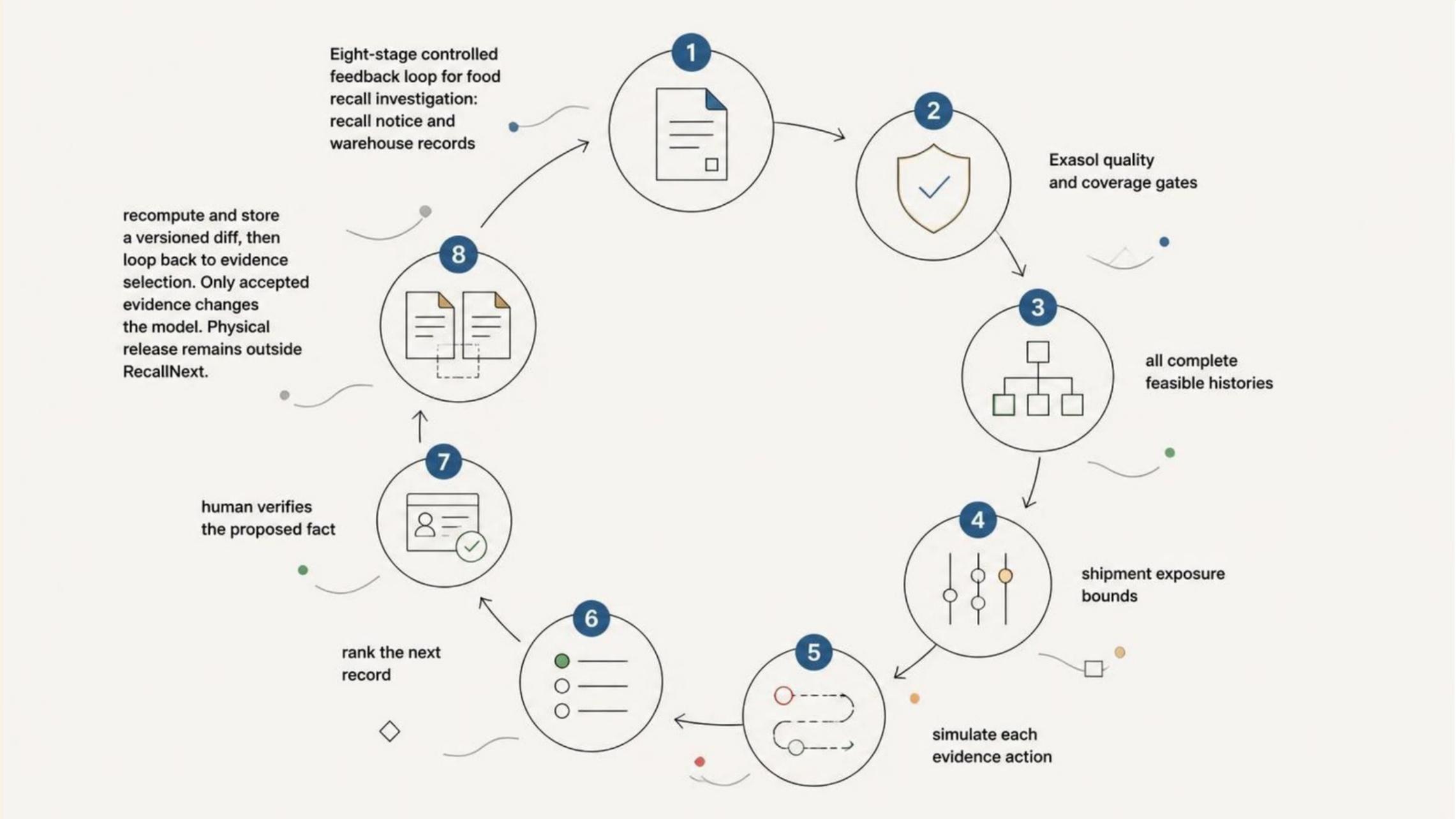
The missing layer between traceability and action

Traceability records what happened. RecallNext tests which missing fact could change what happens next.

EPCIS / TRACEABILITY	Records events and relationships	Gap: does not select the next missing fact
RECALL WORKFLOW TOOLS	Coordinate recall tasks	Gap: limited scenario-based narrowing
BI DASHBOARDS	Summarize known data	Gap: no evidence-action ranking
DOCUMENT AI	Extract facts from documents	Gap: extracted facts still require verification
RECALLNEXT	Models possible histories and simulates evidence outcomes	Bounded prototype; validation with real warehouse workflows remains necessary.

RecallNext turns a missing record into a controlled feedback loop

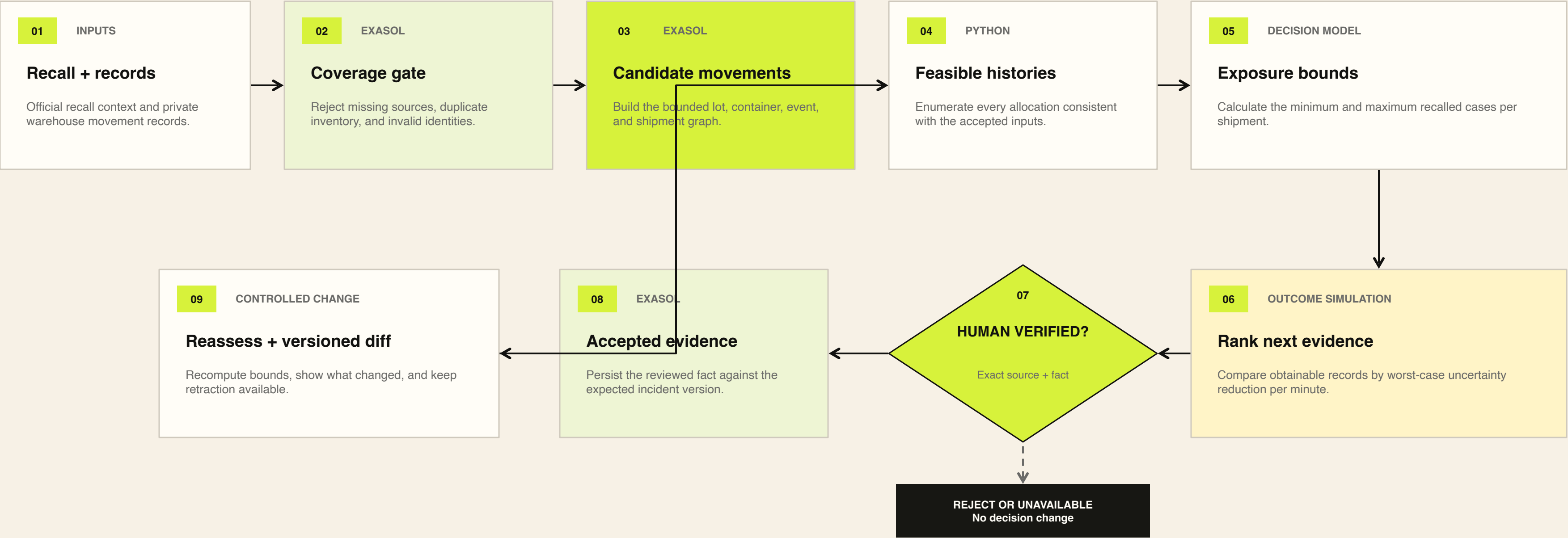
Only accepted evidence changes the model. Physical release remains outside RecallNext.



Evidence selection → verification → recomputation → versioned diff → evidence selection.

How RecallNext turns uncertainty into a controlled decision

Data quality gates protect the model. Human verification controls every evidence-driven change.

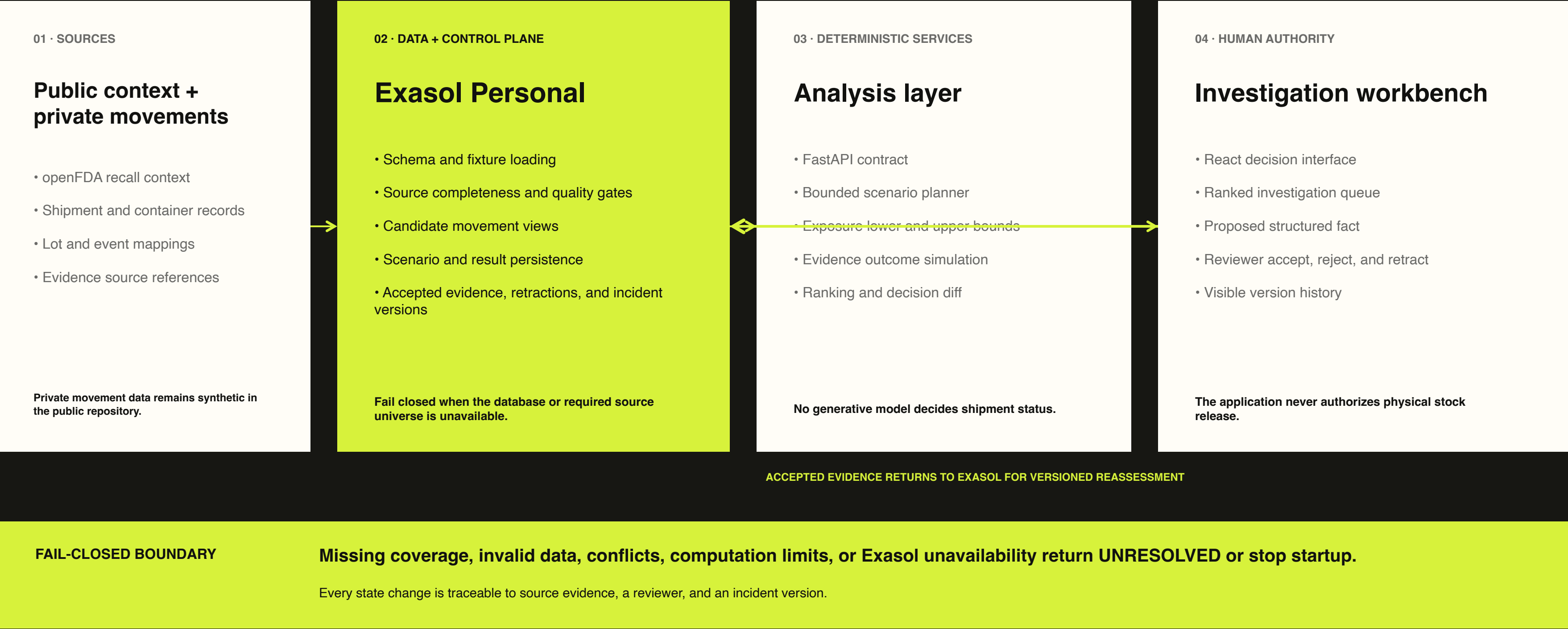


The loop continues until uncertainty is resolved or the available evidence cannot narrow it safely.

Physical release remains outside RecallNext.

Architecture separates data, reasoning, and authority

Exasol owns the durable data and control plane. Deterministic services calculate; the reviewer authorizes evidence.



Selection uses outcome simulation, not a chatbot guess

Rank evidence by the cases it could resolve in the worst valid outcome, not by intuition.

SHIPMENT EXPOSURE

Lower bound → upper bound

Minimum recalled cases across all feasible histories → maximum recalled cases across all feasible histories.

Unavailable, illegible, and conflicting outcomes are non-narrowing outcomes.

SYNTHETIC RETRIEVAL ESTIMATES

Evidence action	Minutes	Valid outcome
Receiving label	4	10 cases resolved
Dispatch manifest	7	Same conditional benefit

Minutes are synthetic retrieval estimates, not measured employee time.

Four decision states preserve uncertainty

Status communicates what the current finite inputs support, and what they do not.

CONFIRMED_INCLUSION

Every feasible history contains at least one recalled case in the shipment.

POSSIBLE_INCLUSION

Some feasible histories contain recalled cases and some do not.

EXCLUDED_UNDER_ASSUMPTIONS

No successful feasible history contains the recalled lot under the recorded finite inputs.

UNRESOLVED

Missing coverage, invalid data, conflict, infeasibility, or a computation limit prevents narrowing.

SAFETY LINE

Excluded under assumptions does not mean safe to consume or automatically released.

Exasol carries the data and control plane

INPUTS

Official openFDA context
Private warehouse records

Warehouse movement data is
synthetic.

CONTROL PLANE

Exasol

Recall, lot, container, event, shipment, source
coverage, scenarios, accepted evidence,
retractions, incident versions.

SERVICES

Bounded Python scenario planner
FastAPI service
React investigation interface

Proof: visible API ran in fail-closed **EXASOL_PERSONAL** mode. If Exasol is unavailable, startup fails instead of silently switching to CSV.

Reviewed evidence returns to Exasol. The interface proposes; human review accepts.